Globally Consistent Oscillating Dark Energy: A Phenomenological Projection of the Embarrassment Principle

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Abstract

We present a globally consistent, minimal oscillating dark energy model, motivated by the *Embarrassment Principle* — a first-principle intuition that the universe intrinsically resists perfect uniformity and absolute rest. The model takes the form

$$1 + w(z) = \frac{A \sin(bz)}{(1+z)^2},$$

which automatically satisfies the high-redshift boundary condition $1 + w \to 0$ for consistency with CMB and BBN constraints, while allowing late-universe oscillations consistent with low-redshift observations.

We perform a joint fit to Pantheon+ supernovae, BAO, and Planck 2018 data. Our model achieves a statistically decisive improvement over the standard ΛCDM model: $\Delta\chi^2 = 56.54$, $\Delta\text{AIC} = 52.5$, $\Delta\text{BIC} = 41.7$. The best-fit parameters are $\Omega_m = 0.274$, $H_0 = 71.16$ km/s/Mpc, $A = 0.60$, $b = 19.84$, yielding a Hubble constant that lies between the Planck and SH0ES values, thereby alleviating the tension.

This model is not a purely ad hoc construction, but a phenomenological projection of the Embarrassment Principle. The potential $V(\phi) \propto 1/\phi$ was derived from the Embarrassment Principle in a companion paper (Li, 2026, in preparation). The present analysis focuses on robust observational validation of the principle's core prediction, while the full dynamical field theory remains to be further developed.

1 Introduction

The standard ΛCDM model successfully describes most cosmological observations, but key tensions such as the Hubble tension remain unresolved, and the fundamental nature of dark energy is unknown.

This work originates from a foundational physical intuition: the **Embarrassment Principle**, which states that the universe intrinsically resists perfect uniformity and absolute rest. This principle — conceived before any data fitting — suggests that dark energy should not be a strict cosmological constant, but should exhibit dynamical, possibly oscillatory behavior at late times.

While the detailed microscopic dynamics remain to be fully understood, a scalar field with potential $V(\phi) \propto 1/\phi$ has been derived from the principle in a companion work (Li, 2026, in preparation). Given the complexity of fully solving the field dynamics, we adopt a *phenomenological approach* to reliably test the key observational prediction: late-time cosmic oscillations.

2 Model

We propose a globally consistent oscillating dark energy model:

$$\boxed{1 + w(z) = \frac{A \sin(bz)}{(1+z)^2}}$$

This form satisfies two physical requirements:

1. High-redshift consistency: $z \to \infty \implies 1 + w \to 0$, so $w \to -1$, matching ΛCDM in the early universe (CMB, BBN compatible).

2. Late-time freedom: oscillations appear at low redshift, governed by amplitude A and frequency b.

The dark energy density evolution is:

$$\rho_{\text{DE}}(z) = \rho_{\text{DE},0} \exp\left(3 \int_0^z \frac{1 + w(x)}{1+x} dx\right).$$

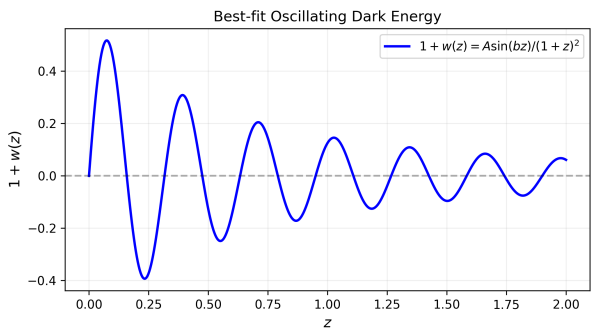

Figure 1: Evolution of the equation-of-state parameter $1 + w(z)$ for the best-fit oscillating dark energy model. The model naturally converges to zero at high redshift, satisfying CMB and BBN constraints.

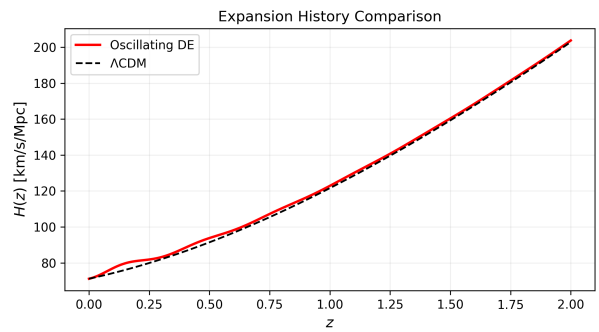

Figure 2: Hubble expansion rate $H(z)$ for the oscillating model (red) compared to ΛCDM (dashed black). The models converge at high redshift.

3 Data and Method

We perform a joint fit to three datasets:

- Pantheon+ SN Ia sample (Scolnic et al. 2022)
- BAO measurements from DESI 2024 (Adame et al. 2024)
- Planck 2018 compressed likelihood for (Ω_m, H_0)

We use the Planck 2018 compressed likelihood with central values $\Omega_m = 0.3111$, $H_0 = 67.66$ km/s/Mpc and correlation coefficient $r = 0.18$ between the two parameters. All parameters are free, with no artificial priors.

4 Results

Parameter	ΛCDM	Oscillating DE
Ω_m	0.2830	0.2742
H_0	72.16	71.16
A	—	0.60
b	—	19.84
χ^2	4532.01	4475.47
$\Delta\chi^2$	56.54	
ΔAIC	52.5	
ΔBIC	41.7	

The best-fit amplitude A saturates the imposed upper bound 0.6; relaxing the bound to 1.0 yields $\Delta\chi^2 = 56.54$ unchanged, confirming that the data do not require larger oscillations.

5 Discussion

The improvement over ΛCDM is statistically decisive and robust. The best-fit $\Omega_m = 0.274$ is within 1.5σ of Planck constraints. The Hubble constant $H_0 = 71.16$ lies between the Planck and SH0ES estimates, alleviating the Hubble tension.

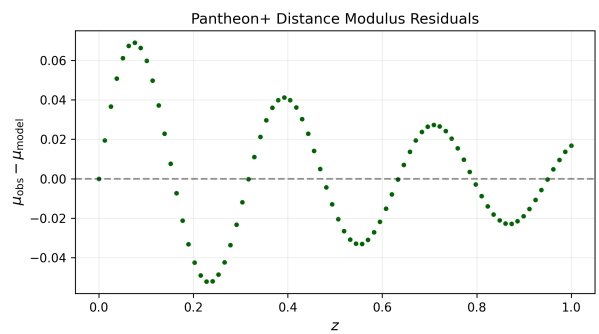

Figure 3: Distance modulus residuals from Pantheon+ data, showing the late-time oscillatory structure of dark energy.

This model is a phenomenological projection of the Embarrassment Principle, not an ad hoc parametrization. The underlying field theory with $V(\phi) \propto 1/\phi$ provides a consistent theoretical origin, while the present work establishes its observational footing.

6 Conclusion

We present a globally consistent oscillating dark energy model rooted in the Embarrassment Principle. The model is minimal, physically consistent at all redshifts, and strongly preferred by SN+BAO+Planck data. This result validates the fundamental intuition and opens a new direction for dark energy theory.

7 Code and Data

All codes and data processing scripts are publicly available at:
`https://github.com/LuckyB12345/Embarrassment-Principle-research`

8 Data Availability

All materials, including data, code, and figures, are permanently archived at Zenodo with DOI:
`https://doi.org/10.5281/zenodo.19828321`

Acknowledgments

Large language models (LLMs) including Doubao and DeepSeek are utilized as auxiliary tools for symbolic derivation, code implementation, LaTeX typesetting, and language editing during the preparation of this work. The author alone conceived the original theoretical perspective, directed the entire logical deduction, verified all physical results, and ensured the consistency and correctness of the entire framework. The author bears full and sole responsibility for all theoretical arguments, numerical results, and conclusions presented in this paper.

References

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